

Devang Jagdish Jalag

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Professional Summary

Master of Applied Computer Science student with hands-on experience designing AWS-native serverless backends via Terraform IaC, building Python and Java distributed systems, and shipping full-stack React/Flask applications integrating LLM APIs. Comfortable across the SDLC: defense-in-depth IAM design, automated testing, peer code reviews, and incident-aware deployment. Actively seeking a Fall 2026 Software Engineering Co-op.

Education

Dalhousie University

Sep 2025 - Dec 2026 (Expected)

Master of Applied Computer Science (MACS) | CGPA: 4.1/4.3

L.D. College of Engineering (GTU)

Oct 2021 - May 2025

Bachelor of Engineering, Computer Engineering | CGPA: 8.87/10 | **Advanced Coursework:** 50-Hr Image Processing & CV (IIT Gandhinagar PMRF)

Skills

Languages: Python, Java, JavaScript

Cloud & Infrastructure: AWS Lambda, AWS S3, AWS API Gateway (HTTP v2), AWS CloudFront, AWS Cognito, AWS IAM, AWS ECS Fargate, AWS EventBridge, AWS CloudWatch, Origin Access Control (OAC), Terraform, Firebase Storage

Databases: AWS DynamoDB, PostgreSQL, Firebase Firestore

Frameworks & Libraries: Flask, SQLAlchemy, boto3, Leaflet.js, Google Gemini 2.5 Flash API, google-genai SDK, bleach, BeautifulSoup, PyJWT, Jackson JSON, jline, expo-location

Work Experience

Palm Infotech | Software Developer Intern

Jan 2025 - Apr 2025

Surat, India

- Engineered a full-stack mobile lottery application, improving UI rendering speed and session retention, by developing a cross-platform Flutter frontend integrated with GetStorage and advanced image caching techniques.
- Accelerated lottery matching logic and result processing workflows, reducing data retrieval latency by ~25%, by architecting robust NestJS REST APIs and implementing complex MongoDB aggregation pipelines alongside MySQL schemas.
- Strengthened system architecture and API request validation, ensuring stable data handoffs across backend services, by implementing NestJS Data Transfer Objects (DTOs), validation pipes, and secure authentication flows.

Academic Projects

CloudNotes | Cloud Engineering

Jan 2026 - Apr 2026

Dalhousie University | **Tech:** AWS Lambda, DynamoDB, Terraform, CloudFront

- Engineered a direct-to-S3 image upload pipeline using SigV4 pre-signed PUT URLs generated by Lambda, bypassing API Gateway's 10 MB payload limits.
- Implemented defense-in-depth access control across 3 layers via Terraform IaC using CloudFront OAC, API Gateway HTTP v2 JWT authorizer, and least-privilege IAM execution roles.
- Built an immutable note version-history system that snapshots full note state to DynamoDB, capped at MAX_VERSIONS via Lambda env var, paired with a UUID-hex share-token system.

DalParkAid | Full-Stack

Jan 2026 - Apr 2026

Dalhousie University | **Tech:** React Native, Firebase, Algorithms, Expo

- Built a three-signal parking prediction engine fusing time/weather heuristics, crowdsourced reports weighted by exponential freshness decay, and a 42-day historical pattern model.
- Enforced submission data quality using haversine-formula proximity validation (30m geofence) and hardened cloud uploads via Firebase Storage security rules.
- Maintained cross-client consistency across four concurrent Firestore onSnapshot subscriptions using atomic read-modify-write transactions for community vote counts.

Interactive World Map | Frontend & Infrastructure

Mar 2026

Dalhousie University | **Tech:** Leaflet.js, JavaScript, Terraform, CloudFront

- Enforced zero-trust S3 access via CloudFront Origin Access Control (OAC) with SigV4 signing — all public bucket ACLs blocked in Terraform.
- Implemented real-time GeoJSON layer re-styling on every keystroke with a 3-state style resolver computed in O(1) via JS Map lookups.
- Configured deterministic 30-color palette assignment with etag cache-busting in Terraform to ensure zero stale-asset serving on deployments.

Dalhousie University | **Tech:** AWS Fargate, Cognito, S3, EventBridge

- Chained API Gateway's native JWT authorizer against Amazon Cognito with application-layer claim extraction, enforcing per-user data isolation in DynamoDB.
- Implemented defense-in-depth S3 access control by enabling all Block Public Access settings, AES256 server-side encryption, and 3600-second presigned URL expiry.
- Designed an ephemeral ECS Fargate janitor triggered by EventBridge scanning DynamoDB for expired entries and issuing targeted S3 deletions.

DalOverflow | *Full-Stack / AI*

Sep 2025 - Dec 2025

Dalhousie University | **Tech:** Flask, PostgreSQL, Google Gemini API, Security

- Integrated Google Gemini 2.5 Flash API for AI-powered Q&A answer generation and multi-answer summarization with truncation-aware two-pass retry.
- Built layered stored-XSS defense using Python bleach with a regex pre-filter stripping injection; enforced at model creation and PUT edits via SQLAlchemy hooks.
- Designed a reusable JWT Flask decorator supporting dual API/session auth routing; applied optimistic concurrency control on question edits returning HTTP 409 Conflict.

Distributed DBMS Prototype | *Backend*

Sep 2025 - Dec 2025

Dalhousie University | **Tech:** Java, Distributed Systems, Concurrency, Sockets

- Implemented a coordinator-worker distributed DBMS in Java supporting range-based row partitioning and vertical column partitioning across 2 worker nodes.
- Engineered distributed atomicity with 2PC-style coordination across all participant workers per DML operation using ReentrantLock-guarded execution.
- Built a custom multi-client TCP database server with wound-wait deadlock detection, per-session Thread dispatch, and a B+ tree index layer for range queries.

Certificates

AWS Solutions Architect – Associate (In Progress), AWS Cloud Foundations, GCP, IBM SkillsBuild Data & AI, Image Processing & CV (IIT Gandhinagar PMRF)